

MM711
Programming for Analysis with SAS

CASE STUDY ANALYSIS

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1. INTRODUCTION

In today's real estate market, property prices are no decided by guesswork, instead they are based on data and statistical analysis. This study is done by combining two datasets, House Price and Characteristics to understand which factor affect the property values.

The main aim of this analysis is to find which factor influences the house price and to build a model to predict the price. The main focus of the study is on factors such as living area, No. of bedrooms, No. of bathrooms and the year the house built. The process includes data preparation, cleaning, exploring, visualization and regression analysis which turns raw data into meaningful insights that help to understand the housing market.

2. DATA PREPARATION AND CLEANING

2.1 Data Import and Merging

Two data sets were imported to SAS through **PROC IMPORT** procedure. The house prices data set consisted of 210 observations on four variables (Reference, Construction Year, Year sold, Sale Price), while the housing characteristic data set consisted of 200 observations on five variables (Reference, Living Area (sm), Garage (sm), Bedroom, Bathrooms, Garage Type). Two data sets were combined based on the key variable "reference" using the SAS merge procedure after PROC SORT on both data sets. After merging, there were 194 observations remaining in the resulting data set. Another variable "House Age" was calculated from YearSold -Construction Year

2.2 Missing Value

Following the merging, missing values were assessed separately for the house price variables and the housing characteristic variable using **PROC MEANS** with **NMISS** option.

For the housing characteristics dataset, the PROC MEANS output from **Fig.1** confirmed that there are no missing values observations existed for any of the five variables: Reference, LivingArea_sm, Garage_sm, Bedroom, and Bathroom all had 0 missing values. Thus, the housing characteristics file was a complete one.

For the house price dataset, output from **Fig.2** confirmed that two variables have missing values. Since sale year is needed to calculate the age of the house and SoldPrice is the dependent variable in the regression equation, these cases were removed from the analysis, leading to an analytic sample size of 194 observations.

Garage type was considered separately using **PROC FREQ**. Of the 200 records in the characteristics file, 158 had recorded garage type. From **Fig. 3**, we can see that 52.53% of the properties has attached garage, 32.91% has detached garage and 14.56% were recorded having no garage (NA). It is noted that 42 observation (21%) had a missing value for garage

type. Significant number of missing values Garage_type was excluded from the list of predictors in regression analysis

Variable	N Missing	Action Taken
YearSold	3	Excluded from analysis
SoldPrice	1	Excluded from analysis
Garage_Type	42 (21%)	Excluded from the model; Garage_sm
All Characteristics variable	0	Retained

3. Distribution of Sale Prices

Before modelling, the distribution of the dependent variable, SoldPrice was examined using **PROC UNIVARIATE** with normality test **Fig.3** and a Q-Q plot **Fig.4**.

Test	Statistic	p-value	Conclusion
Shapiro-Wilk	W=0.9958	p=0.8732	Normal
Kolmogorov-Smirnov	D=0.0323	p>0.15	Normal
Cramer-von Mises	W ² =0.0289	p>0.25	Normal
Anderson-Daling	A ² =0.1932	p>0.25	Normal

Since all the normality tests showed p-values greater than 0.05, we can say with confidence that the SoldPrice variable is normally distributed. This is an assumption of OLS regression. Since this assumption holds true for our variable, there was no need to transform it using any method such as taking the log of the variable.

From **Fig.5** the **PROC TTEST** analysis confirmed a mean sale price of £106,155 (95% CI: £102,057 to £110,253) with a standard deviation of £28,938. Both **Fig.6** and **Fig.7** demonstrated a symmetric bell-shaped distribution without any significant skewness or heavy tailed data, which can be explained based on the normality test results. In addition, the total price variation was found to lie between £28,000 to £176,000.

4. Correlation Analysis

Pearson correlation coefficients were computed between SoldPrice and the four potential continuous predictors (LivingArea_sm, Bedroom, Bathroom, HouseAge) using **PROC CORR**.

VARIABLES	PEARSON r WITH SOLDPRICE	INTERPRETATION
LivingArea_sm	r=0.660	Strong positive correlation
Bathroom	r=0.536	Moderately positive correlation
HouseAge	r=-0.613	Moderately strong negative correlation
Bedroom	r=0.251	Weak positive correlation

As shown in **Fig.8**, the one with highest correlation coefficient is the living area ($r=0.660$). thus, the more spacious property is, the more expensive its selling price. It is an obvious finding in real estate economy. The number of bathrooms was positively correlated ($r=0.536$). The reason for that is more bathrooms add comfort, making properties more attractive and increases the market value. The house age was negatively correlated ($r=-0.613$). Older houses tend to sell for less price, as the buyers prefer to buy new properties with modern features and low-maintenance. The analysis shows negative and non-statistically significant relationship between the No. of bedrooms and price($r=0.251$) suggesting that bedrooms doesn't influence the price.

5. Visualization and Distribution

As illustrated in **Fig.9**, the scatter diagrams a positive linear relationship between LivingArea_sm and SoldPrice shows upward sloping trendline. **Fig.10**, indicates that house price generally increases from one to three bedrooms, with little additional increase beyond that point, reflecting the weak Pearson correlation observed in the dataset.

6. Multiple Linear Regression Model

6.1 Model Specification

A multiple linear regression model was fitted using **PROC REG** to predict SoldPrice from five independent variables: LivingArea_sm, Garage_sm, Bedroom, Bathroom, and HouseAge. The model takes the form:

$$\text{SoldPrice} = \beta_0 + \beta_1(\text{LivingArea_sm}) + \beta_2(\text{Garage_sm}) + \beta_3(\text{Bedroom}) + \beta_4(\text{Bathroom}) + \beta_5(\text{HouseAge}) + \varepsilon$$

where β_0 is the intercept, β_1 – β_5 are the regression coefficients for each predictor, and ε is the error term. All 194 complete observations were used in estimation.

6.2 Overall Model Fit

The **ANOVA table** from **Fig.11** shows that the model was statistically significant at ($F(5, 188) = 101.18, p < 0.0001$). This means that the combination of the five predictors has significantly explained a large amount of variation in the sale price of the houses. In this case, the value of the coefficient of determination was 0.7291. This means that about 72.91% of the total variation in the sale prices of the houses has been explained by the five predictors. The adjusted R-squared statistic was 0.7219, suggesting that the model can be said to be generalizable.

From the regression analysis result, the root mean square error (RMSE) was estimated at 15,262. With an average sale price of £106,155, this means that the standard error can be said to be 14.4%.

6.3 Parameter Estimates

The individual regression coefficients significance related to their p-value are presented in the table below and the standard errors, t-statistics are in **Fig.12**

VARIABLE	P-VALUE	SIGNIFICANCE
Intercept	<0.0001	Significant
LivingArea_sm	<0.0001	Significant
Garage_sm	<0.0001	Significant
Bedroom	0.6231	Not Significant
Bathroom	0.0093	Significant
HouseAge	<0.0001	Significant

7. Research Question

Does the Age of the house Negatively affect its market value?

Variables Used:

Independent Variable – HouseAge

Dependent Variable – SoldPrice

A multiple Linear Regression model was used to analyze the relationship between house age and sold price. House age was included as one of the predictor variables in the regression analysis. The regression analysis shows a negative relationship between HouseAge and SoldPrice. The regression coefficient for HouseAge was -392.49 with a statistically significant p-value less than 0.0001. For every year the selling price decreases by approximately 392 units by keeping all other variables constant. This means house age has an important influence on property value.

8. Model Diagnostics

A comprehensive set of diagnostic plots was produced by PROC REG to assess whether the model assumptions of linearity, homoscedasticity, normality of residuals, and absence of influential outliers were met.

8.1 Residual Distribution

From **Fig.13**, the histogram of residuals indicates a symmetrical distribution with its mean approximately equal to zero. The Q-Q plot of the residuals revealed that the standardized residuals were close to the straight line on almost the whole range of residuals except for the ends. Both the graphs confirm the assumption about the normality of residual distribution.

8.2 Residuals vs Predicted Values

The scatterplot for the residuals plotted against the fitted values displayed no particular pattern, indicating that the residuals were scattered randomly about zero over the entire range of fitted values. This validates the assumptions of linearity and homogeneity of variance, apart from the minor widening in the middle range.

8.3 Residuals by Individual Regressors

The residual-by-regressor plots from **Fig.14** didn't reveal any clear non-linear relationship between any predictor and the response variables. Residuals for LivingArea_sm, Garage_sm, Bathroom and HouseAge were randomly spread out over zero along their respective ranges of variation, which confirms that linear regression fits the model well for each predictor. In the case of bedroom, residuals were spread out for all four levels of variation, which is in line with the significance of this predictor.

9. Discussion

The analysis aimed at identifying the influence of a certain set of structural and temporal characteristics of properties on their sale price. The multiple linear regression is carried out for 194 real estate deals accounted for about 72.9% of variation in sale price (adjusted $R^2=0.7219$). This is a very good result for such cross-sectional real estate sample. The whole model proved to be statistically highly significant ($F = 101.18$, $p < 0.0001$).

Four of the five variables used were statistically significant predictors of the price at which the house was sold. Living Area exhibited a strong positive correlation with the dependent variable, where an increase in living area by one square meter resulted in an increase in the price by £617. The size of the garage was also positively associated with price, contributing £401 per square meter to the cost. The age of the house exhibited a strong negative correlation with house price, reducing the value by £392 per year. Number of bathrooms positively influenced the property prices by £5,524 per bathroom.

In contrast, the number of bedrooms which was found to be statistically insignificant considering other attributes, this could probably be due to its strong correlation with square footage. Tests conducted showed that the regression met the required assumption with

normally distributed and heteroskedastic errors. However, missing Garage_Type data limited further analysis of garage categories.

All diagnostic test showed that the model is correctly specified. The residuals are normally distributed; heteroscedasticity was not detected and showed no non-linear patterns against any of the regressors. No individual observation has an unduly large impact on the estimated results. There is only one potential issue regarding the existence of 42 missing observations of Garage_type (accounting for 21% of characteristics information), which prevented the utilization of Garage_type as a dummy variable, since the effect of attached/detached provision can be investigated further using the whole set of variables.

10. Conclusion

In conclusion, the analysis demonstrates that living area, garage size, house age and the number of bathrooms determine the price at which houses are sold within this dataset. This study has produced a sound model to predict the house sale price based on the available variables. Future work could explore the inclusion of location variables, condition ratings, and energy performance data to further improve predictive accuracy.

References

- Malpezzi, S. (2003). Hedonic pricing models: A selective and applied review. In T. O'Sullivan & K. Gibb (Eds.), *Housing Economics and Public Policy*. Oxford: Blackwell.
- Rosen, S. (1974). Hedonic prices and implicit markets: Product differentiation in pure competition. *Journal of Political Economy*, 82(1), 34-55.

APPENDIX A – SAS Code

/*Importing the Dataset*/

```
Filename Reffile"/home/u64454765/MM711 Programming with  
SAS/Data/Characteristics.xlsx";
```

```
proc import datafile=Reffile
```

```
    DBMS=XLSX
```

```
    Out=Char;
```

```
    Getnames=Yes;
```

```
Run;
```

```
Filename Reffile"/home/u64454765/MM711 Programming with  
SAS/Data/Houseprice.xlsx";
```

```
proc import datafile=Reffile
```

```
    DBMS=XLSX
```

```
    Out=HousePrice;
```

```
    Getnames=Yes;
```

```
Run;
```

/*Remove Duplicates*/

```
proc sort data=HousePrice out=prices_unique nodupkey;
```

```
    by Reference;
```

```
run;
```

```
proc sort data=char out=chars_sorted;
```

```
    by Reference;
```

```
run;
```

/*Checking Missing Values*/

```
proc means data=char nmiss;
```

```
title "Missing Values in Housing Characteristics";
```

```
Run;
```

```
proc means data=HousePrice nmiss;
```

```
title "Missing Values in House Price";
```

```
run;
```

```

proc freq data=Char;
title "Missing Garage Types";
table Garage_type;
Run;

/*Merging & Cleaning the Datasets*/
data house_clean;
    merge chars_sorted(in=a) prices_unique(in=b);
    by Reference;
    if Garage_Type = "NA" then Garage_Type = "";
    if SoldPrice = . or YearSold = . then delete;
    if LivingArea_sm <= 0 or LivingArea_sm > 500 then delete;
    HouseAge = YearSold - ConstructionYear;
run;

/*Removing Extreme Values for Final Data*/
data Final;
set house_clean;
HouseAge = YearSold - ConstructionYear;
where livingArea_sm lt 1000;
Run;

/* Test for normality */
proc univariate data=WORK.FINAL normal mu0=0;
    ods select TestsForNormality;
    var SoldPrice;
run;

/* t test */
proc ttest data=WORK.FINAL sides=2 h0=0 plots(showh0);
    var SoldPrice;
run;

```

/*Correlation Analysis*/

```
ods noproctitle;  
ods graphics / imagemap=on;  
proc corr data=WORK.FINAL pearson nosimple noprob plots=none;  
    var LivingArea_sm Bedroom Bathroom HouseAge;  
    with SoldPrice;  
run;
```

/*Visualization*/

```
proc sgplot data=Final;  
    title "Impact of Living Area on Sold Price";  
    scatter x=LivingArea_sm y=SoldPrice;  
    reg x=LivingArea_sm y=SoldPrice / degree=1; /* Adds a trend line */  
run;
```

```
proc sgplot data=Final;  
    title "Sold Price Distribution by Number of Bedrooms";  
    vbox SoldPrice / category=Bedroom;  
run;
```

```
proc sgplot data=Final;  
    title "Distribution of Property Sale Prices";  
    histogram SoldPrice;  
    density SoldPrice; /* Adds a distribution curve */  
run;
```

/*Regression*/

```
proc reg data=Final;  
    title "Multiple Linear Regression: Predicting Sold Price";  
    model SoldPrice = LivingArea_sm Garage_sm Bedroom Bathroom HouseAge;  
run;  
quit;
```

APPENDIX B – RESULT IMAGE

Missing Values in Housing Characteristics

The MEANS Procedure

Variable	Label	N Miss
Reference	Reference	0
LivingArea_sm	LivingArea_sm	0
Garage_sm	Garage_sm	0
Bedroom	Bedroom	0
Bathroom	Bathroom	0

FIG. 1

Missing Garage Types

The FREQ Procedure

Garage_Type				
Garage_Type	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Attached	83	52.53	83	52.53
Detached	52	32.91	135	85.44
NA	23	14.56	158	100.00
Frequency Missing = 42				

FIG. 2

Missing Values in House Price

The MEANS Procedure

Variable	Label	N Miss
Reference	Reference	0
ConstructionYear	ConstructionYear	0
YearSold	YearSold	3
SoldPrice	SoldPrice	1

FIG.3

Missing Garage Types

The UNIVARIATE Procedure
Variable: SoldPrice (SoldPrice)

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.995812	Pr < W	0.8732
Kolmogorov-Smirnov	D	0.032344	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.02886	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.193232	Pr > A-Sq	>0.2500

FIG.4

The TTEST Procedure

Variable: SoldPrice (SoldPrice)

N	Mean	Std Dev	Std Err	Minimum	Maximum
194	106155	28938.4	2077.7	28000.0	176000

Mean	95% CL Mean	Std Dev	95% CL Std Dev
106155	102057 110253	28938.4	26316.9 32144.4

DF	t Value	Pr > t
193	51.09	<.0001

FIG.5

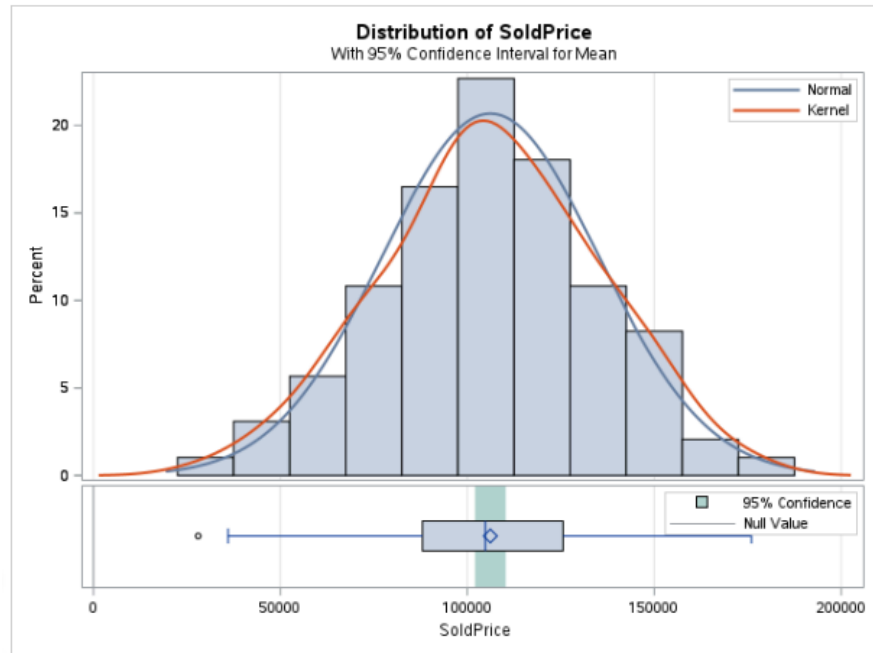


FIG.6

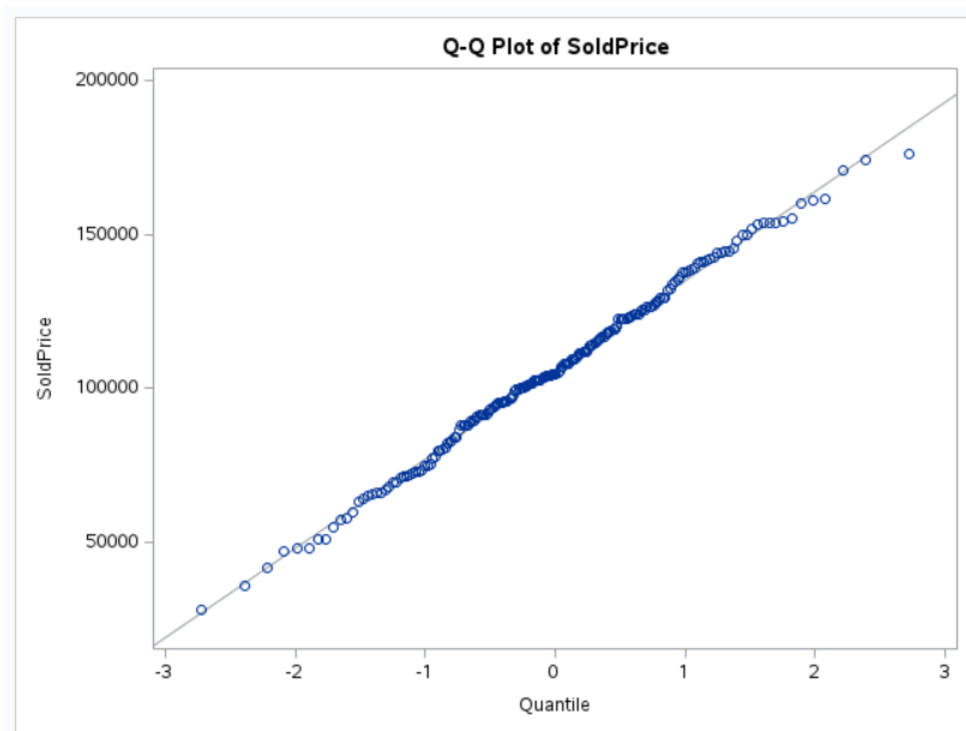


FIG.7

1 With Variables:	SoldPrice
4 Variables:	LivingArea_sm Bedroom Bathroom HouseAge

Pearson Correlation Coefficients, N = 194				
	LivingArea_sm	Bedroom	Bathroom	HouseAge
SoldPrice	0.65981	0.25139	0.53629	-0.61332
SoldPrice				

FIG.8



FIG.9

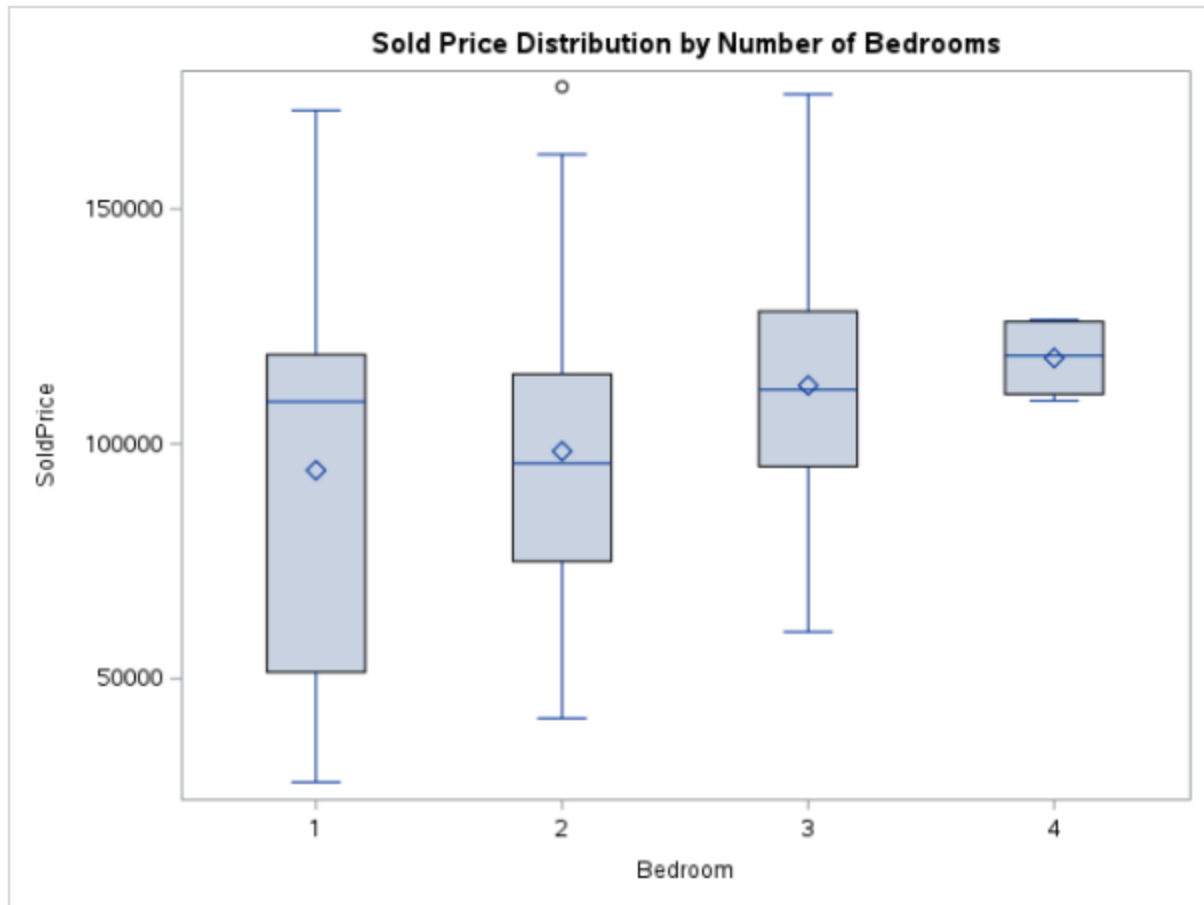


FIG.10

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	1.178357E11	23567147949	101.18	<.0001
Error	188	43787900169	232914363		
Corrected Total	193	1.616236E11			

Root MSE	15262	R-Square	0.7291
Dependent Mean	106155	Adj R-Sq	0.7219
Coeff Var	14.37664		

FIG.11

Parameter Estimates						
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	Intercept	1	40785	6876.19883	5.93	<.0001
LivingArea_sm	LivingArea_sm	1	616.68374	66.49532	9.27	<.0001
Garage_sm	Garage_sm	1	400.54387	74.57952	5.37	<.0001
Bedroom	Bedroom	1	-968.85039	1967.84940	-0.49	0.6231
Bathroom	Bathroom	1	5523.92888	2103.08227	2.63	0.0093
HouseAge		1	-392.49170	48.14723	-8.15	<.0001

FIG.12

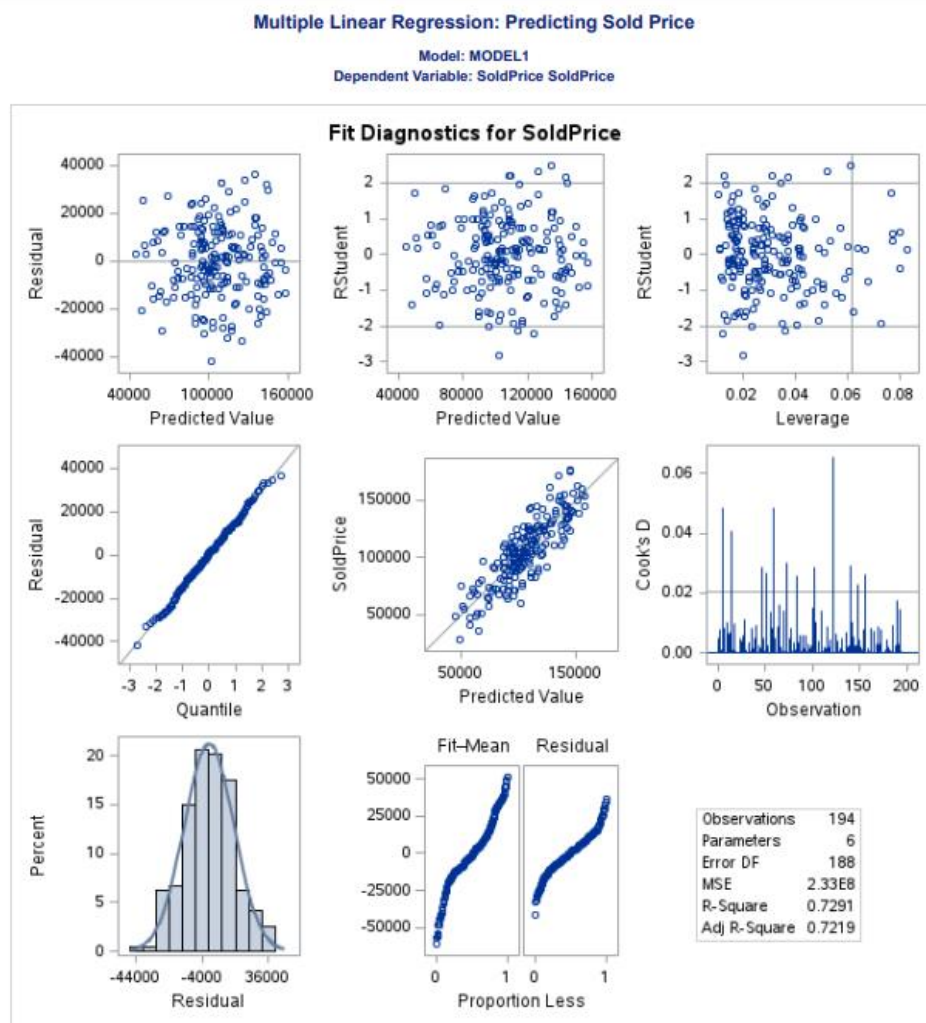


FIG.13



FIG.14